

Vidyut Prajna (Electric Intelligence)

Theme 9: AI for EV Charging Optimization & Infrastructure Planning
by BESCOM

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1. Executive Summary

As Bengaluru accelerates its Electric Vehicle (EV) transition, BESCOM faces the **Coincident Peak Challenge**: the high-probability overlap of vehicle charging with existing residential/commercial peak windows. Unmanaged charging creates localized thermal stress on distribution transformers (DTRs), threatening grid stability.

Vidyut Prajna is a non-invasive "sidecar" decision-support layer. It treats the city as a living energy organism, using **Graph-based Neural Networks** to predict demand and **Lagrangian Optimization** to balance load. Our solution ensures that the EV revolution supports, rather than stresses, the Bengaluru grid.

2. Problem Understanding & Constraints

- Grid failures often occur at the 11kV/440V transformer level even when city-wide load is stable.
- Demand is non-uniform; evening residential spikes in HSR Layout differ fundamentally from daytime logistics surges in Peenya.
- The system operates as a data-driven overlay with **zero modification** to existing SCADA or hardware systems.

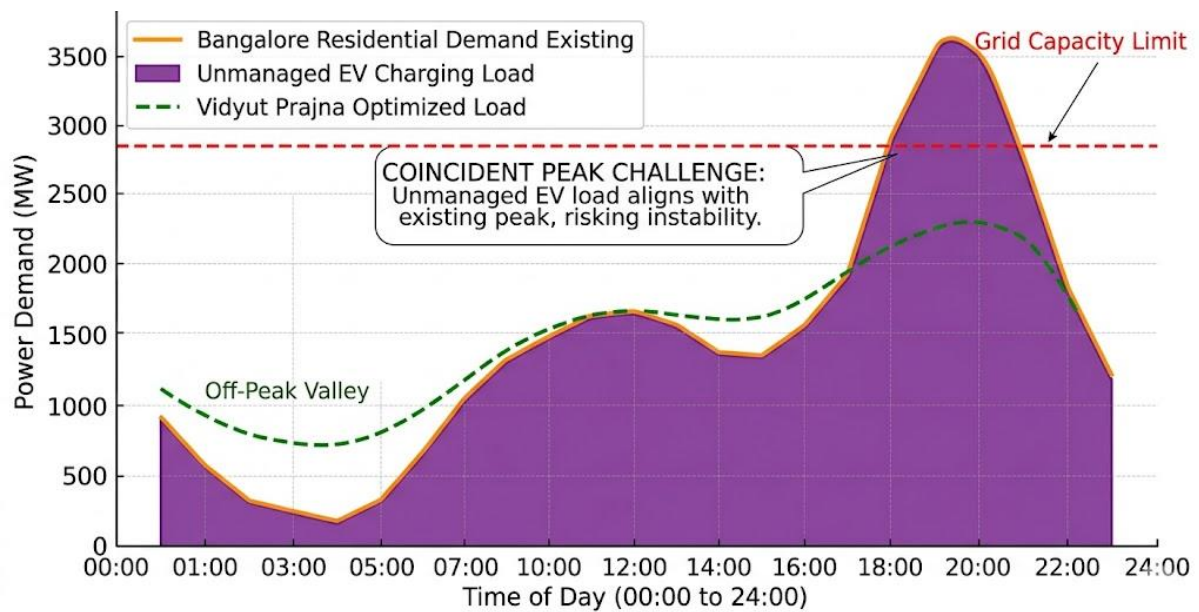


Figure 1 Impact of Vidyut Prajna: Shifting coincident unmanaged EV demand (solid orange) to the optimized night-time valley (dashed green).

3. Methodology: The Technical Engine

3.1 Spatio-Temporal Graph Convolutional Networks (STGCN)

We partition Bengaluru into hexagonal cells using **Uber's H3 Index (Level 9)**.

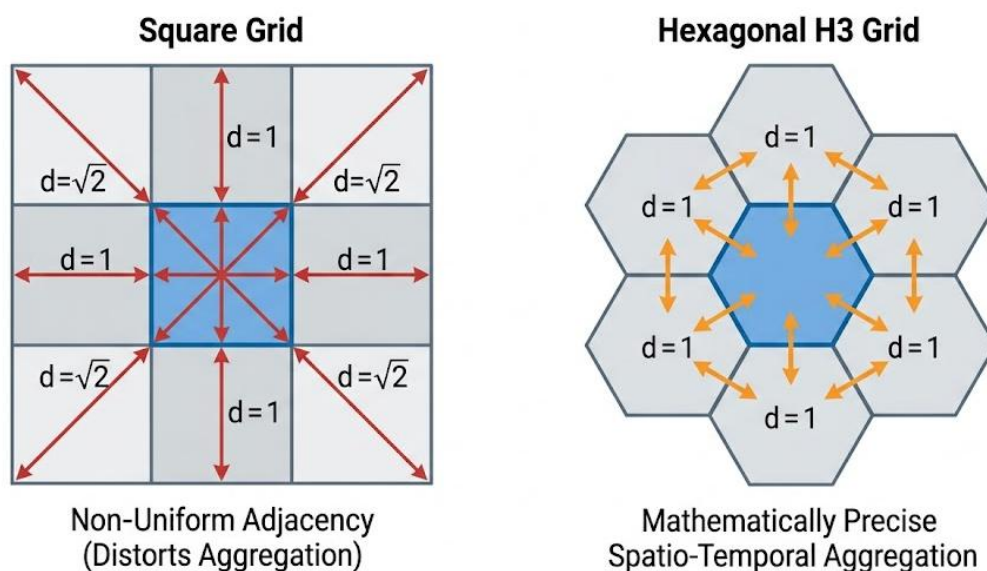


Figure 2 The Isotropic Advantage: Unlike square grids with varying diagonal distances, H3 hexagons guarantee uniform neighbor distances, crucial for accurate Kernel Density Estimation (KDE).

- Standard models ignore city topology. STGCN learns how a traffic bottleneck on the Outer Ring Road acts as a leading indicator for a charging spike in an adjacent cell 20 minutes later.

- Unlike square grids, hexagons have equal distances between all neighbours, which is mathematically vital for accurate **Kernel Density Estimation (KDE)** of vehicle arrivals.

3.2 Constraint-Aware Scheduling (The Optimizer)

We implement a **Lagrangian-based Multi-Criteria Decision Making (MCDM)** engine.

- We minimize the global grid variance

$$J = \alpha \cdot \sum_t (P_{grid(t)} - P_{avg})^2 + \beta \cdot \sum_i (\omega_i \cdot \Delta SoC_i)$$

Where α minimizes load variance and β ensures high-priority vehicles ω_i meet their State of Charge ΔSoC_i targets.

- The system searches for the "Point of Minimum Regret" balancing grid health against user service levels.

4. Infrastructure Siting Logic

Our siting engine identifies "Future-Proof" corridors by fusing three critical data layers:

- Mapping congregation points for 2W/3W delivery fleets (Swiggy/Zomato) the primary drivers of near-term demand.
- Real-time and historical spare capacity in local transformers.
- Hexagonal cells where demand growth outpaces current charger density within a 500m radius.

5. System Architecture: The Isolation Layer

The architecture is designed for **Zero-Risk deployment**.

- Ingests telemetry (OCPP, traffic, weather, grid) using **Apache Kafka** with K-anonymity masking for privacy.
- Processes data through the STGCN and Lagrangian optimizer.
- Surfaces actionable insights (ranked zones, suggested windows, and feeder impact summaries) to the Planner Dashboard.
- No direct commands are sent to distribution hardware, satisfying the safety-first constraint.

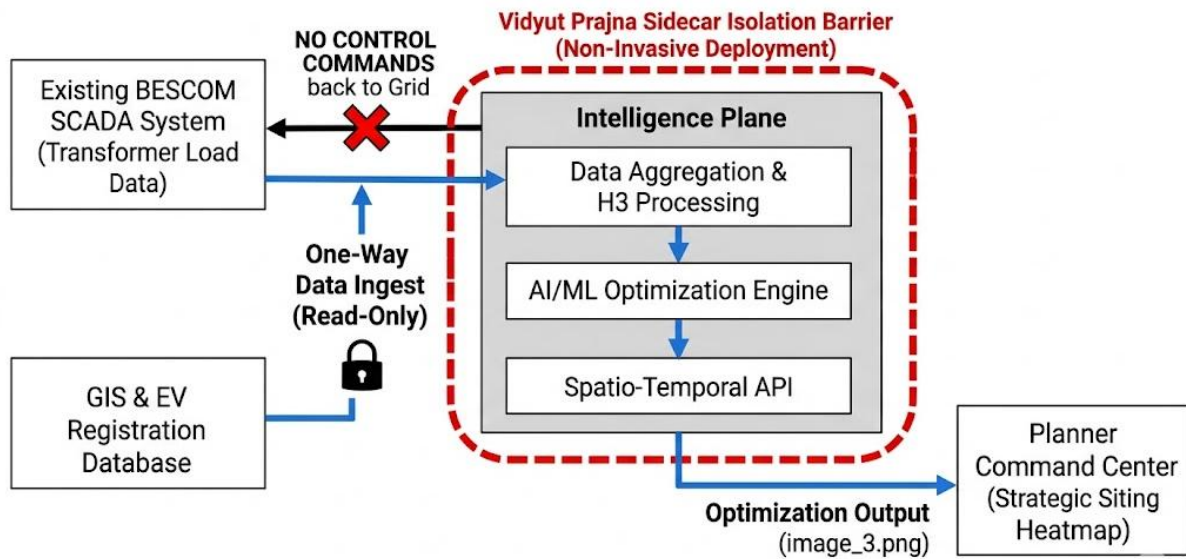


Figure 3 Non-Invasive Sidecar Architecture: A dashed isolation layer separates the Apache Kafka data stream and Lagrangian optimizer from BESCOM's grid control systems.

6. Implementation Roadmap

We move from a "Codebase" to a "City-scale" solution through a staged rollout.

Phase 1: Concept	Graph & Logic	Finalize the mathematical "Shadow Pricing" for load shifting.
Phase 2: Prototype	Digital Twin	Launch a single-ward (e.g., Whitefield) twin using Shadow Mode to stress-test the optimizer.
Phase 3: Pilot	Planner Command Center	Full-scale deployment of the dashboard for BESCOM planners to identify 2027 infrastructure needs.

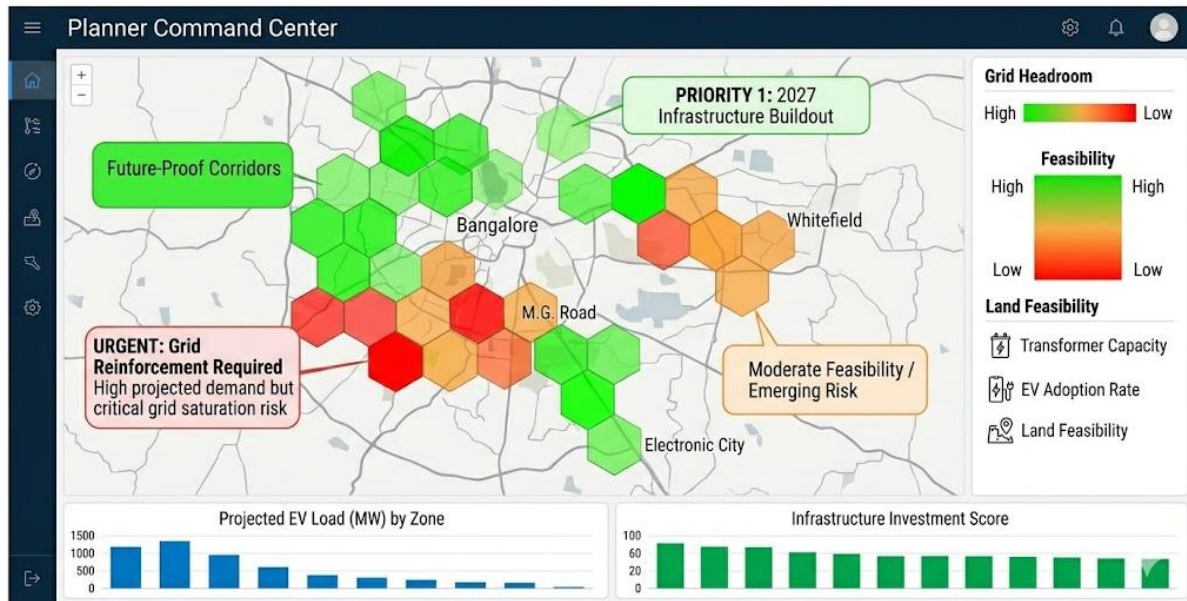


Figure 4 The Strategic Planner Dashboard: A GIS-scale command center visualizing optimized siting recommendations, highlighting high Feeder Headroom (green) and Siting Risk (red).

7. Evaluation & Baselines

Success is measured by the Delta against unmanaged systems:

- Targeted **15-20% reduction** in predicted transformer overload events.
- Lower **Peak-to-Average Ratio (PAR)**, reducing the need for expensive "peaker-plant" power procurement.
- Improvement in charger utilization rates compared to **Uniform Placement** strategies.

8. Conclusion

Vidyut Prajna treats EV charging as a strategic planning problem, not just a technical feature. By providing an explainable, non-invasive intelligence layer, we empower BESCOM to lead the national transition to green mobility without compromising the reliability of the city's power grid.